

Do CRA Ratings Guarantee Fair Lending? Evidence from Mortgage Approvals and Pricing in the South Atlantic

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Abstract

This paper examines whether the Community Reinvestment Act (CRA) effectively promotes equitable lending outcomes. Using the 2022 Home Mortgage Disclosure Act (HMDA) data matched with CRA performance ratings, I investigate disparities in both mortgage approval rates and loan pricing across CRA-regulated and non-CRA lenders in the South Atlantic region. Logistic regression results show that while CRA lenders exhibit higher overall approval rates, they are significantly less likely to approve low-income applicants compared to non-CRA institutions, contradicting the core objective of the CRA. Furthermore, OLS analysis of interest rate spreads reveals that certain racial groups, particularly Hispanic and Native Hawaiian borrowers, face higher pricing penalties from CRA lenders. These patterns indicate that CRA ratings may signal the quantity of lending, but not necessarily the fairness of lending. The findings underscore the need for the CRA to reform loan pricing and borrower-level outcomes into its assessment framework.

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1 Introduction

Ensuring equitable access to credit is a central challenge in the U.S. financial system. The Community Reinvestment Act (CRA), enacted in 1977, was designed to combat redlining and encourage depository institutions to meet the credit needs of low- and moderate-income (LMI) communities. CRA evaluations are a critical part of the U.S. regulatory landscape, with high ratings often interpreted as a signal that a bank is fulfilling its obligations to serve underserved communities (Office of the Comptroller of the Currency (OCC), 2018). However, whether a strong CRA rating translates into truly fair lending outcomes, particularly for economically disadvantaged and minority borrowers, remains an open question.

Existing literature has primarily focused on loan access, investigating whether CRA institutions are more likely to approve mortgage applications than their non-CRA counterparts.

While early studies focused on whether CRA increased loan origination in underserved communities (Avery et al., 2003; Bhutta, 2011), more recent work questions whether CRA obligations affect loan pricing and borrower outcomes beyond mere approval rates (Ringo, 2017; Ding and Nakamura, 2017). However, the connection between CRA obligations and pricing disparities remains underexplored, which this paper seeks to address. This omission is critical because disparities in pricing can perpetuate financial inequality even when approval rates are high.

This paper aims to fill this gap by examining whether CRA ratings are associated not only with higher loan approval rates but also with more equitable loan pricing. Using loan-level data from the 2022 Home Mortgage Disclosure Act (HMDA) merged with CRA performance evaluations, I investigated two key questions.

- (1) Do CRA-regulated lenders approve mortgage loans at higher rates compared to non-CRA lenders?
- (2) Do CRA-regulated lenders offer better loan pricing - measured by the interest rate spread relative to the average prime offer rate (APOR) - particularly for historically underserved groups?

The analysis focuses on the South Atlantic region of the United States, which provides a large, diverse data set that covers more than 930,000 mortgage applications. This setting offers a valuable case study for understanding how CRA impacts lending practices in a racially and economically diverse context.

The results reveal a complex picture. On the surface, CRA lenders exhibit higher overall ap-

proval rates relative to non-CRA lenders, consistent with the law’s intended effect. However, this benefit does not extend uniformly across borrower groups. Low-income applicants are paradoxically less likely to be approved by CRA lenders, suggesting that banks subject to CRA may adopt more conservative underwriting for economically disadvantaged borrowers, possibly due to concerns over credit risk and regulatory scrutiny. Furthermore, the pricing analysis reveals that racial disparities persist even among approved loans. Specifically, Hispanic, Native Hawaiian, and some Asian subgroups face significantly higher interest rate spreads at CRA institutions compared to non-CRA lenders.

These findings contribute to the growing literature questioning whether current CRA regulations adequately promote financial equity. While CRA ratings may capture efforts to expand credit access, they appear insufficient for ensuring fairness in the terms of that credit. This underscores an important policy implication: the CRA’s evaluation framework should move beyond assessing loan quantity to include metrics related to loan pricing and borrower-level outcomes. Failure to do so risks perpetuating structural inequalities under the guise of regulatory compliance. In summary, this paper provides empirical evidence that CRA ratings, as currently structured, are an imperfect proxy for fair lending. Addressing this shortcoming will require both enhanced regulatory standards and more granular public disclosure of lending practices by income and race. The goal is not merely to increase the number of loans but to ensure that those loans are offered on equitable terms to all borrowers.

2 Data and Methods

2.1 Data Sources and Sample Construction

This study draws on loan-level data from the 2022 Home Mortgage Disclosure Act (HMDA), which provides comprehensive records of mortgage applications submitted to financial institutions in the United States. The HMDA dataset includes information on loan outcomes (approved, denied), applicant demographics (race, ethnicity, gender, income), loan characteristics (amount, type, term, purpose), and pricing variables (interest rates and rate spreads).

To capture the regulatory treatment of lenders, the HMDA data is merged with publicly available Community Reinvestment Act (CRA) performance evaluation data, obtained from the Federal Financial Institutions Examination Council (FFIEC). Lenders are classified as CRA-regulated if they are depository institutions subject to CRA obligations; non-CRA lenders are primarily independent mortgage companies or fintech firms not subject to CRA supervision. The merger uses the Legal Entity Identifier (LEI) as the matching key.

Additionally, benchmark interest rates are matched from the Average Prime Offer Rate (APOR) dataset, published by the Federal Financial Institutions Examination Council. This allows for the calculation of interest rate spreads, which measure the pricing premium of each loan relative to market standards controlling for loan term and origination year.

The analysis is restricted to mortgage applications within the South Atlantic region, which includes nine states (Delaware, Florida, Georgia, Maryland, North Carolina, South Carolina, Virginia, West Virginia, and the District of Columbia). This region offers both demographic diversity and significant variation in lender types, making it an ideal setting for evaluating the impact of CRA obligations on lending outcomes. After excluding applications with missing key variables (e.g., loan amount, applicant income, race, pricing information), the final sample consists of approximately 932,000 mortgage applications.

2.2 Key Variables

The empirical analysis centers on two primary outcome variables: loan approval and loan pricing. Loan approval is defined as a binary indicator equal to one if the loan application was approved and originated, and zero if the application was denied. Loan pricing is captured by the interest rate spread, calculated as the difference between the loan’s reported annual percentage rate (APR) and the benchmark Average Prime Offer Rate (APOR), conditional on loan term and origination year. This measure reflects the degree to which a loan is priced above the prevailing market rate, serving as a proxy for potential pricing disparities and the presence of high-cost loans.

The key explanatory variable is the lender’s CRA regulatory status, coded as one for lenders subject to CRA obligations and zero for lenders that are not. This classification distinguishes depository institutions, which are regulated under the CRA, from independent mortgage companies and fintech lenders that are exempt.

To examine income-based heterogeneity, the analysis includes an indicator for low-income applicants, defined as borrowers whose household income falls below 80 percent of the area median income (AMI), consistent with CRA’s regulatory focus on serving low- and moderate-income communities.

Racial and ethnic group membership is captured through categorical indicators derived from HMDA’s self-reported data. The main analysis employs broad racial categories, including White, Black or African American, Asian, Hispanic or Latino, and Native Hawaiian or Other Pacific Islander. Disaggregated subgroups are further examined in robustness checks to assess heterogeneity within broader racial categories.

Interaction terms are constructed to evaluate whether the association between CRA regulation and lending outcomes varies systematically across applicant income levels and racial groups. Specifically, the analysis includes interaction terms between CRA status and (1) low-income status and (2) racial or ethnic group membership.

A comprehensive set of control variables is included to adjust for loan, borrower, and market characteristics. Loan-level controls comprise the logarithm of loan amount, loan term, loan purpose (purchase versus refinance), loan type (conventional versus government-backed), and loan product features such as open-end lines of credit, balloon payments, and interest-only payments. Borrower-level controls include gender, credit score type (as reported), and a neighborhood-level economic indicator measuring the ratio of tract median income to metropolitan statistical area (MSA) median income. Additionally, the models incorporate geographic fixed effects at the state and MSA levels to account for unobserved heterogeneity in local credit market conditions.

Table 1: Variable Definitions

Variable	Definition	Type
Loan Approval	1 if loan application approved and originated; 0 if denied	Binary (Outcome)
Interest Rate Spread	APR minus APOR (matched by loan term and year)	Continuous (Outcome)
CRA Status	1 if lender is CRA-regulated; 0 otherwise	Binary (Key)
Low-Income Applicant	1 if applicant income $\leq$ 80% of area median income	Binary (Key)
Race/Ethnicity	White, Black, Asian, Hispanic, NHPI, Others	Categorical (Key)
CRA $\times$ Low-Income	Interaction of CRA Status and Low-Income Applicant	Interaction Term
CRA $\times$ Race/Ethnicity	Interaction of CRA Status and Race/Ethnicity	Interaction Term
Log Loan Amount	Natural logarithm of loan amount	Continuous (Control)
Loan Term	Loan term in months	Continuous (Control)
Loan Purpose	Purchase vs. Refinance	Categorical (Control)
Loan Type	Conventional vs. Government-backed	Categorical (Control)
Loan Features	Open-end line, Balloon payment, Interest-only payment	Binary Indicators (Control)
Gender	Applicant and Co-applicant gender	Categorical (Control)
Credit Score Type	Type of credit score used	Categorical (Control)
Neighborhood Income Ratio	Tract median income divided by MSA median income	Continuous (Control)
Geographic Fixed Effects	State and MSA indicators	Fixed Effects

2.3 Empirical Strategy

The empirical analysis examines whether CRA-regulated lenders differ systematically from non-CRA lenders in both mortgage approval decisions and loan pricing outcomes. The identification strategy leverages cross-sectional variation in lender CRA status, conditional on a rich set of borrower, loan, and geographic characteristics. While the assignment of CRA regulation is not random, the inclusion of detailed controls aims to mitigate confounding to the extent possible. Consequently, the results are interpreted as conditional associations rather than causal effects.

2.3.1 Loan Approval Model

To estimate differences in loan approval probabilities, the following logistic regression model is specified:

$$\begin{aligned} \text{logit}(\text{Approval}_i) = & \beta_0 + \beta_1 \cdot \text{CRA}_i + \beta_2 \cdot \text{LowIncome}_i + \beta_3 \cdot \text{Race}_i \\ & + \beta_4 \cdot (\text{CRA} \times \text{LowIncome})_i + \beta_5 \cdot (\text{CRA} \times \text{Race})_i \\ & + \mathbf{X}_i + \delta_g + \epsilon_i \end{aligned} \quad (1)$$

Where:

- *Approval* is a binary indicator equal to one if the loan is approved and originated.
- *CRA* indicates whether the lender is CRA-regulated.
- *LowIncome* indicates whether the applicant's income is below 80% of the area median income.
- *Race* is a vector of categorical indicators for applicant race/ethnicity.
- *CRA* $\times$ *LowIncome* and *CRA* $\times$ *Race* are interaction terms capturing heterogeneous CRA effects.
- $\mathbf{X}_i$ is a vector of loan and borrower controls (e.g., loan amount, loan term, loan type, loan purpose).
- δ_g denotes geographic fixed effects at the state and MSA level.
- ϵ_i is an idiosyncratic error term.

The primary coefficients of interest are β_1 , which captures the baseline difference in approval rates associated with CRA regulation, and the interaction terms β_4 and β_5 , which assess

whether CRA effects vary systematically by income status and race.

2.3.2 Loan Pricing Model

For originated loans, an ordinary least squares (OLS) regression is employed to examine disparities in loan pricing, measured by the interest rate spread:

$$\text{Spread}_i = \alpha_0 + \alpha_1 \text{CRA}_i + \alpha_2 \text{Race}_i + \alpha_3 (\text{CRA} \times \text{Race})_i + \alpha_4 \text{Approved}_i + \mathbf{X}_i + \delta_g + \epsilon_i \quad (2)$$

Where:

- Spread_i is the interest rate spread for loan i , defined as the APR minus the corresponding APOR benchmark.
- Approved_i controls for whether the loan was approved, accounting for potential selection effects based on underwriting criteria.
- Other variables are defined as in Equation (1).

The coefficient α_1 captures the average difference in loan pricing between CRA and non-CRA lenders, while α_3 assesses whether this difference varies across racial or ethnic groups. The inclusion of Approved_i adjusts for possible differences in borrower risk profiles that are associated with the approval decision.

2.4 Robustness and Sensitivity Checks

As part of the empirical strategy, I also conduct robustness checks to assess the stability of the results. Specifically, regressions are estimated separately for CRA and non-CRA lenders to examine whether borrower characteristics have differential impacts under different regulatory regimes. In addition, I perform group-wise t-tests comparing approval rates and interest spreads between CRA and non-CRA lenders. To further explore racial and ethnic disparities, approval rate gaps are calculated across racial groups. Finally, an XGBoost model with SHAP-based feature importance is employed to validate the consistency of the findings under a flexible, non-linear framework.

2.5 Limitations

This study is subject to several important limitations inherent to observational research designs. First, CRA regulatory status is not randomly assigned; lenders subject to CRA may systematically differ from non-CRA lenders along unobserved dimensions, including institu-

tional risk preferences or underwriting policies. Although the inclusion of detailed borrower, loan, and geographic controls mitigates observable confounding, unobserved heterogeneity may still bias the estimates.

Second, the cross-sectional nature of the data constrains the ability to make dynamic or longitudinal inferences about the impact of CRA regulation over time. Additionally, the HMDA dataset lacks key borrower-level risk indicators such as credit scores, debt-to-income ratios, and loan-to-value ratios, which may limit the precision of risk adjustment in the pricing models.

Consequently, the results should be interpreted as capturing conditional associations rather than causal relationships. While the findings provide important insights into the relationship between CRA regulation, credit access, and loan pricing disparities, they do not establish definitive causal claims.

3 Results

3.1 Loan Approval Disparities

Table 2 presents the results from the logistic regression model examining the relationship between CRA regulation and loan approval outcomes. The baseline specification indicates that CRA-regulated lenders are associated with a higher overall probability of loan approval relative to non-CRA lenders (coefficient = 0.0568, $p < 0.01$). This finding aligns with the intended policy function of the CRA, which is to expand credit access, particularly in underserved communities.

The coefficient for Low-Income is large, negative, and highly significant (coefficient = -0.6605, $p < 0.01$), indicating that low-income applicants face substantially lower odds of loan approval. More importantly, the interaction term between CRA status and Low-Income is also negative and statistically significant (coefficient = -0.9908, $p < 0.01$). This result suggests that CRA-regulated lenders are even less likely to approve loans for low-income applicants compared to non-CRA lenders. This counterintuitive finding implies that, conditional on observables, CRA institutions may apply more stringent underwriting standards to low-income borrowers, potentially due to risk management concerns or regulatory compliance incentives.

The racial effects are also pronounced. Applicants identified as Black (coefficient = -0.7050, $p < 0.01$), NHPI (coefficient = -0.7196, $p < 0.01$), and Asian Indian (coefficient = -0.6608, $p < 0.01$) all face significantly lower odds of loan approval relative to White applicants. In contrast, Filipino applicants have a large positive coefficient (-1.0452, $p < 0.01$), suggesting

a lower likelihood of loan approval. These results highlight substantial racial disparities in loan approval outcomes, even after controlling for loan characteristics and geography.

Overall, the model confirms that while CRA status is associated with higher general approval rates, this benefit does not extend to low-income applicants. Moreover, significant racial disparities persist within the lending process.

Table 2: Logit Regression Results for Loan Approval

Variable	Coef.	Std. Error
CRA Status	0.0568***	0.012
Low-Income	(0.6605)***	0.012
Race: Black	(0.7050)***	0.054
Race: NHPI	0.7196***	0.112
Race: Asian Indian	(0.6608)***	0.098
Race: Philipino	(1.0452)***	0.273
CRA $\times$ Low-Income	(0.9908)***	0.036
Controls	Yes	
Geographic Fixed Effects	Yes	
Observations	58,2209	
Pseudo R-squared	0.1443	

Note: Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

3.2 Loan Pricing Disparities

Table 3 reports the OLS regression results for interest rate spreads. The results reveal that CRA-regulated lenders are associated with substantially higher loan pricing. Specifically, the coefficient on CRA Status is 93.92 basis points and statistically significant at the 1% level, indicating that, holding all else constant, loans issued by CRA lenders carry an average spread nearly 94 basis points higher than those from non-CRA lenders.

Approval status is positively associated with the interest spread (coefficient = 4.71, $p < 0.01$), suggesting that borrowers whose loans are approved tend to receive higher spreads, potentially reflecting compensations for riskier profiles or unobserved factors not captured in the model.

The racial patterns are particularly noteworthy. While the main effects for racial categories are generally small and not statistically significant, the interaction terms between CRA status and race are large, negative, and highly significant across all groups. For example, the

CRA $\times$ Black interaction coefficient is -72.17, and CRA $\times$ NHPI is -91.32, both significant at the 1% level. This indicates that although CRA lenders charge higher overall spreads, minority borrowers experience substantially lower spreads within the CRA-regulated segment compared to non-CRA lenders. Similar patterns hold for Asian (-87.99) and White (-83.89) applicants.

These large negative interaction terms suggest that the baseline CRA premium (93.92 bps) is partially offset for minority borrowers. However, given that the main CRA status coefficient is extremely large, it is possible that while racial disparities are mitigated under CRA lenders, the overall pricing remains higher relative to non-CRA lenders.

Loan characteristics also have meaningful effects. The log of loan amount has a positive and highly significant coefficient (3.87), indicating that larger loans are associated with higher spreads. This may reflect risk-based pricing or underwriting policies that scale with loan size. Neighborhood income is negatively associated with interest spreads (coefficient = -0.0102, $p < 0.01$), suggesting that loans issued in higher-income neighborhoods tend to have slightly lower spreads.

Overall, the results confirm that CRA status is a major driver of loan pricing disparities, and that racial pricing gaps are significantly conditioned by whether the lender is CRA-regulated. While CRA lenders charge higher average spreads, the pricing penalty for minority borrowers is mitigated within the CRA segment.

Table 3: OLS Regression for Interest Rate Spread

Variable	Coef.	Std. Error
CRA Status	93.9164***	(10.021)
Race: Black or African American	-1.5606	(5.372)
Race: NHPI	-2.8453	(8.843)
Race: Asian	-1.3720	(5.686)
Race: White	-5.9219	(5.345)
Has CRA $\times$ Race: Black	-72.1719***	(10.143)
Has CRA $\times$ Race: NHPI	-91.3156***	(17.024)
Has CRA $\times$ Race: Asian	-87.9955***	(10.534)
Has CRA $\times$ Race: White	-83.8866***	(10.042)
Approval Status	4.7139***	(0.593)
Log Loan Amount	3.8721***	(0.417)
Neighborhood Income Ratio	-0.0102***	(0.001)
Controls	Yes	
Geographic Fixed Effects	Yes	
Observations	59,3033	

Note: Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

4 Robustness and Sensitivity Checks

To validate the robustness of the main findings, several complementary analyses are conducted.

First, I estimate separate regressions for CRA-regulated lenders and non-CRA lenders to examine whether borrower characteristics have differential effects under different regulatory regimes. Table 4 shows that while certain patterns hold consistently across both groups, substantial divergences also emerge. For example, the effect of being Black or African American is positive and significant in the CRA lender sample, suggesting that CRA lenders reduce racial disparities in approval rates. In contrast, the effect is strongly negative and significant in the non-CRA sample. Similarly, the NHPI group exhibits a sharp divergence: a negative and significant coefficient under CRA but a positive coefficient under non-CRA lenders, though not always statistically significant. These results reinforce that the relationship between borrower characteristics and loan approval is heterogeneous and conditioned by CRA status.

Table 4: Regression Coefficients by CRA Status: CRA vs Non-CRA Lenders

Variable	CRA Coef.	CRA Sig.	Non-CRA Coef.	Non-CRA Sig.
Black or African American	0.1657	***	-1.0101	***
Asian (General)	0.5468	***	0.2142	***
Asian Indian	0.0552	not	0.2434	***
Vietnamese	-0.3684	not	-0.3684	not
Native Hawaiian	-1.1754	not	1.5185	not
NHPI (General)	-1.5251	***	0.0929	***
Filipino	-0.0475	not	0.1136	not
White	0.5553	***	0.2495	***
Low Income	-0.7334	***	-0.8096	***
Interest-Only Loan	1.0176	***	0.9286	***
Balloon Payment	-0.2878	***	-0.1824	***
Open-End Line	0.2557	***	0.1928	***
Loan Amount	0.0000001554	***	3.028E-07	***
Loan Term	0.0024	***	-0.00005885	***

Note: *** $p < 0.01$. “not” indicates not statistically significant at conventional levels.

Second, a series of t-tests comparing CRA and non-CRA lenders confirms that disparities extend beyond loan approval to loan pricing. As shown in Table 5, CRA lenders charge significantly higher interest spreads on average (53.87 bps) compared to non-CRA lenders (24.07 bps), with a t-statistic of 28.90 ($p < 0.01$). Approval rates also differ substantially between the two groups, with CRA lenders having a higher average approval rate (40.67%) compared to non-CRA lenders (28.89%), as evidenced by a highly significant t-statistic of 100.15.

Table 5: T-Test of Interest Spread and Approval Rate by CRA Status

	CRA	Non-CRA	T-statistic
Interest Spread (percent)	53.87	24.07	28.90***
Approval Rate (bps)	40.67	28.89	100.15***

Note: Spread and Approval Rate is measured in basis points (bps). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Third, I assess racial and ethnic disparities in approval outcomes by calculating the difference in approval rates between CRA and non-CRA lenders for various racial and ethnic groups.

Table 6 highlights that Native Hawaiian and Pacific Islander applicants experience the largest approval gap (-58.9 percentage points) when comparing CRA to non-CRA lenders. Hispanic subgroups also consistently face negative approval gaps, whereas Black applicants experience a positive gap (+22.9 percentage points), indicating improved outcomes under CRA regulation relative to non-CRA lenders.

Table 6: Approval Rate Differences by Race/Ethnicity (CRA – Non-CRA)

Group	Approval Rate Gap
Native Hawaiian or Pacific Islander	-0.589
Samoan	-0.257
Japanese	-0.197
Korean	-0.173
Black or African American	+0.229
Hispanic: Cuban	-0.202
Hispanic: Puerto Rican	-0.149
Hispanic: Other	-0.146
Hispanic: Mexican	-0.116

Lastly, I employ an XGBoost machine learning model with SHAP-derived feature importance to validate the consistency of the regression-based findings. Table 7 shows that CRA status, neighborhood income ratio, race (particularly Black identity), and loan size are among the top predictors of interest rate spreads. This confirms that the disparities observed in loan pricing are not artifacts of linear model specification but are salient across flexible, non-parametric approaches as well.

Taken together, these robustness checks provide strong validation for the main findings. They demonstrate that the disparities identified are not driven by specific modeling choices and that CRA status meaningfully conditions how borrower characteristics translate into lending outcomes.

Table 7: Key Feature Importance from XGBoost Model Predicting Interest Spread

Feature	Importance Score
CRA Status	0.184
Neighborhood Income Ratio (Tract to MSA)	0.126
Log Loan Amount	0.027
Approval Status	0.007
Derived Race: Black or African American	0.021
Derived Race: Race Not Available	0.008
Applicant Sex: Male	0.040
State Code: MD	0.344
State Code: FL	0.039
State Code: NC	0.013

Note: This table reports the top features relevant to CRA status, income, race, loan characteristics, and geographic location. Less relevant technical variables (e.g., credit score types) are omitted for clarity.

5 Discussion

5.1 Summary of Main Findings

This study provides a nuanced assessment of whether CRA regulation leads to more equitable mortgage lending outcomes. The results indicate that while CRA-regulated lenders exhibit higher overall loan approval rates, this benefit does not extend uniformly across borrower groups. Notably, low-income applicants are less likely to be approved at CRA institutions compared to non-CRA lenders, a pattern that runs counter to the intended mission of the CRA.

In terms of loan pricing, CRA-regulated lenders charge higher interest rate spreads on average, particularly disadvantaging certain racial and ethnic groups such as Hispanic and Native Hawaiian applicants. While Black borrowers benefit from higher approval odds at CRA lenders, they still face substantial pricing penalties. These findings suggest that CRA ratings may capture improvements in loan quantity but fail to ensure fairness in loan terms and conditions.

5.2 Potential Mechanisms

Several mechanisms may explain these patterns. One possible explanation is that CRA institutions operate under more stringent regulatory scrutiny, which may incentivize conservative underwriting practices—especially for applicants deemed high-risk, such as low-income borrowers. While CRA compels banks to serve underserved areas, it does not mandate how risk is priced, leading to potential overcompensation via higher interest spreads.

Additionally, non-CRA lenders, including fintechs and mortgage companies, may have greater flexibility in risk assessment and are more willing to approve loans to higher-risk applicants, albeit often at higher costs or with alternative risk-sharing mechanisms. This dynamic helps explain why non-CRA lenders exhibit higher approval rates for low-income borrowers.

Racial pricing disparities may stem from systemic biases embedded in credit risk models, historical underwriting practices, or localized market dynamics. The fact that these disparities persist even after controlling for observable borrower and loan characteristics indicates that CRA, in its current form, may not be sufficient to counteract structural inequities in mortgage lending markets.

5.3 Policy Implications

The findings have significant implications for CRA reform and broader financial regulation. First, CRA evaluations should move beyond measuring loan volume in low- and moderate-income areas to incorporate borrower-level outcomes, including loan pricing fairness. This shift would align CRA assessments more closely with actual financial equity outcomes.

Second, regulators could require more granular disclosure of lending data, particularly interest rate spreads disaggregated by borrower income and race. Enhanced transparency would enable advocacy groups, researchers, and policymakers to monitor and address inequities more effectively.

Third, targeted interventions such as loan assistance programs, refinancing options for high-cost loans, and mandatory fairness audits for CRA-regulated lenders could help mitigate the identified disparities. Addressing pricing discrimination should be as central to CRA’s mission as improving access to credit.

5.4 Limitations and Future Research

This study has several limitations. First, the observational nature of the data constrains causal inference. CRA status is not randomly assigned, and although extensive controls are

included, unobserved heterogeneity may persist. Second, the HMDA dataset lacks critical borrower-level risk metrics such as credit scores, debt-to-income ratios, and loan-to-value ratios, which limits the precision of risk adjustment.

Additionally, the analysis focuses on a single cross-sectional year and a specific geographic region (the South Atlantic), which may limit generalizability. Future research could extend this analysis temporally to assess whether these patterns hold over time or under different regulatory environments. Incorporating richer credit risk data, when available, would also enable more precise identification of the sources of pricing disparities.

Overall, while CRA remains a vital tool for promoting financial inclusion, this study highlights that without substantive reform, it may fall short of delivering true lending equity.

6 Conclusion

This paper investigates whether CRA regulation leads to fairer mortgage lending outcomes in terms of both loan approval and pricing. Using HMDA loan-level data matched with CRA performance ratings for the South Atlantic region, the analysis reveals a complex set of outcomes.

On one hand, CRA-regulated lenders exhibit higher overall loan approval rates, particularly benefiting Black borrowers who experience substantially improved access relative to non-CRA lenders. On the other hand, this benefit does not extend to all groups. Low-income applicants are significantly less likely to be approved at CRA institutions than at non-CRA lenders, contrary to the intended goals of the CRA. Moreover, CRA-regulated lenders charge higher interest rate spreads on average, with pronounced pricing disparities affecting Hispanic, Native Hawaiian, and other minority borrowers.

These findings suggest that while the CRA remains effective in expanding credit access for certain groups, it is insufficient for ensuring equitable lending outcomes in totality. The current regulatory focus on loan quantity, rather than loan quality or fairness in terms, may inadvertently allow pricing inequities to persist or even worsen.

From a policy perspective, this study underscores the urgent need to modernize CRA assessments to include borrower-level outcomes such as loan pricing fairness. Greater transparency in lending data, particularly regarding pricing stratified by race and income, would enable more effective oversight. Furthermore, complementary policy tools—such as targeted loan assistance programs, mandatory fairness audits, and expanded refinancing options for high-cost loans—are critical to closing the gaps that CRA alone does not sufficiently address.

In conclusion, while the CRA continues to play an important role in promoting financial inclusion, this study demonstrates that without significant reform, the CRA’s ability to deliver genuine equity in mortgage lending remains limited. Addressing this challenge requires a shift from measuring lending volume to evaluating the fairness of lending outcomes across both approval and pricing dimensions.

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